

# MAE 6246 – Homework 5

## PBH Tests and Observability

**Name:** \_\_\_\_\_

**Total:** 65 points

**Due:** Friday, October 9, 2026, at 10:00 AM (Eastern Time).

**Submission format.** Submit your analytical work as a single PDF through Blackboard. Your work may be typed or clearly handwritten. Show all calculations and explanations; a computer-algebra system must not replace the analytical work. Both problems are analytical; no code or notebook is required.

**Late policy.** There is a late penalty of **1 point per 10 minutes**. The penalty is enforced strictly except under extraordinary circumstances such as documented illness or emergency.

**Motivation.** Controllability asks which state motions inputs can influence. Observability asks which initial states can be distinguished from known inputs and measured outputs. This assignment connects the PBH tests to these interpretations and examines how actuator and sensor choices affect a system.

### Problem 1 – PBH controllability and actuator choice (30 points)

Consider the continuous-time system  $\dot{x} = Ax + bu$ , where

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad x \in \mathbb{R}^3, \quad u \in \mathbb{R}.$$

The states  $x_2$  and  $x_3$  are two identical first-order channels driven by a common input;  $x_2$  also drives  $x_1$ .

- (a) **Left eigenspace (6 points).** Find the eigenvalues of  $A$  and describe *all* nonzero left eigenvectors  $w^T$  satisfying  $w^T A = \lambda w^T$ . State the dimension of the left eigenspace. Generalized eigenvectors are not required.
- (b) **Single-input PBH test (10 points).** Determine controllability using the rank of  $[\lambda I - A \quad b]$  at each distinct eigenvalue. Find a nonzero left eigenvector with  $w^T b = 0$ . For  $z = w^T x$ , derive the scalar differential equation and its solution. Explain why a nonzero value of this coordinate cannot be reached from  $x(0) = 0$ , regardless of the input.
- (c) **Two independent inputs (10 points).** Replace  $bu$  by  $B_\gamma u$ , where

$$B_\gamma = \begin{bmatrix} 0 & 0 \\ 1 & \gamma \\ 1 & 1 \end{bmatrix}, \quad u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in \mathbb{R}^2, \quad \gamma \in \mathbb{R}.$$

Use the PBH rank test to determine exactly which values of  $\gamma$  make  $(A, B_\gamma)$  controllable. Confirm the result by asking whether any nonzero vector in the left eigenspace satisfies  $w^T B_\gamma = 0_{1 \times 2}$ .

- (d) **Interpretation (4 points).** Explain why checking only the individual vectors of a left-eigenvector basis can give an incorrect conclusion for the single-input system. Also explain how  $x_1$  can be controllable in part (c) even though the first row of  $B_\gamma$  is zero.

## Problem 2 – Observability and sensor choice (35 points)

Consider a different continuous-time system,

$$\dot{x} = Ax + Bu, \quad y = Cx, \quad A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

The input  $u(t)$  is known. Compare the two scalar sensors

$$C_a = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}, \quad C_b = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}.$$

Each sensor measures a sum of two state components. Assume exact, continuous output measurements.

- (a) **Observability matrices (10 points).** For each sensor, construct

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$$

and determine its rank analytically. State whether each pair  $(A, C)$  is observable. Show a determinant, row reduction, or another exact rank argument; a numerical rank alone is insufficient.

- (b) **Observability PBH test (10 points).** Find the right eigenvectors of  $A$  and use the condition  $Av = \lambda v$ ,  $Cv = 0$  to confirm both conclusions from part (a). For each unobservable mode, state its eigenvalue and give a nonzero right eigenvector. Explain why such a vector produces zero output under zero input even though the state need not be zero.
- (c) **Indistinguishable initial states (10 points).** For the unobservable sensor choice, determine  $\ker \mathcal{O}$ . Construct two distinct initial states  $x_0$  and  $\bar{x}_0$  whose outputs are identical for all  $t \geq 0$  under the same known input  $u(t)$ . Derive the state difference and prove that the output difference is zero. Explain why changing the common known input cannot distinguish this pair.
- (d) **Adding a measurement (5 points).** Retain the unobservable sensor and add one measurement of an individual state component. Choose a component that makes the combined sensor observable, write its  $2 \times 3$  output matrix, and verify observability. Briefly explain why one scalar sensor can sometimes reveal all three states over time, although one instantaneous scalar measurement cannot.

**Submission checklist.** Include exact rank arguments, the PBH witness and its interpretation, indistinguishable initial states, and justification of your sensor choice.